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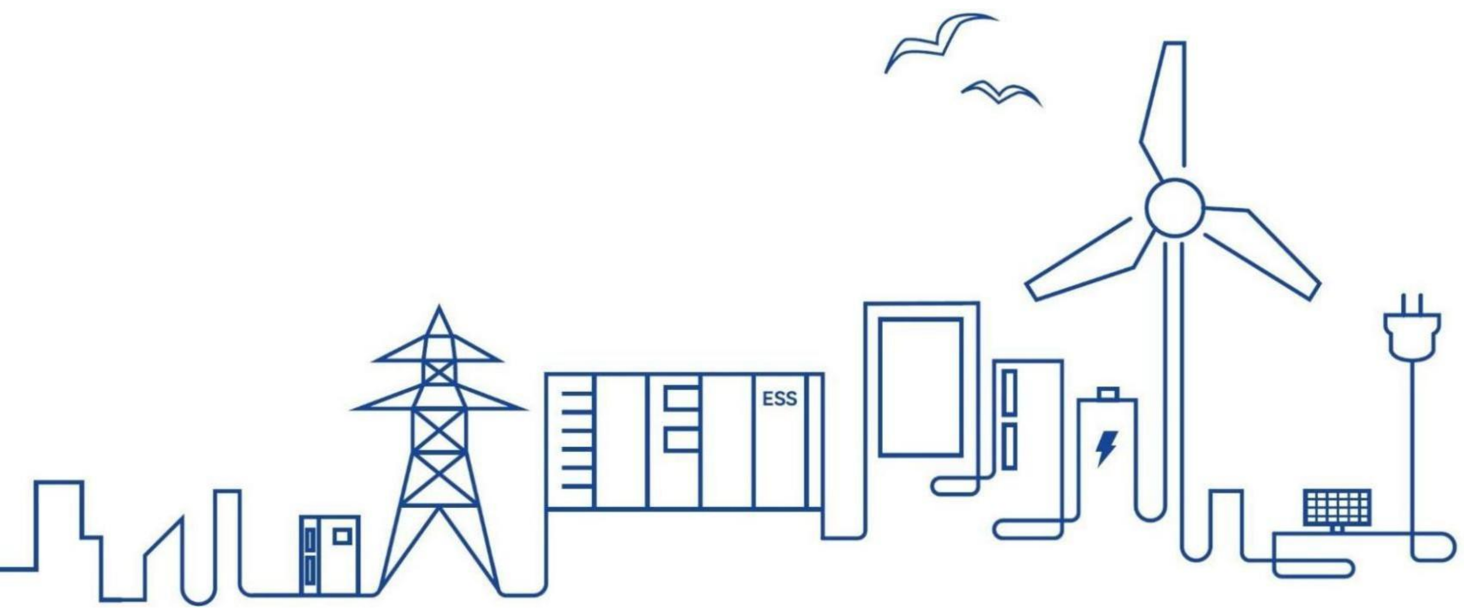
POWER ATLANTIC LIQUID COOLING BATTERY CONTAINER

DATASHEET



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1 Lithium Iron Phosphate Battery Container

1.1 Functions and Features

This container energy storage system is mainly used for storing and transferring energy, which supports battery status monitoring and management, wide application such as grid peak regulation, frequency regulation, grid-connected access of new energy power stations, etc.

This Liquid cooling energy storage system is a highly integrated product, which integrates battery system, battery management system, thermal management system, fire suppression system, power supply and distribution system, monitoring system, environmental control system, and cable routing, etc., and has the characteristics of high safety and reliability, easy installation, high energy density and long cycle life.

1.2 Product Appearance

Figure Appearance of 5.015MWh container (Reference)

1.3 System Description

1.3.1 System Specification

Table Parameters of energy storage system

Item	Specification
Battery capacity (BOL) at DC side	5015kWh

Battery type		LFP
Configuration		12P416S
Rated energy/ kWh		5015
Battery voltage range/ V		1164.8 ~ 1497.6
Duration/ h		≥2
Rated charge/discharge power/current		2500kW/1884A 1250kW/942A
Charging and discharging efficiency		93%
Thermal Management		Liquid cooling
Life cycle (25°C, 90%DOD, 70%SOH)		8000 cycles
Auxiliary power supply		380/400AC,50/60Hz
Self-discharge		<3%permonth (All devices have been turned off and the system has been disconnected)
Dimension (W * D * H)/ mm		6,058 * 2,438 * 2,896
Weight/ T		~ 41.5
Operation temperature/ °C	Charge	0 - 55
	Discharge	-30 - 55
Humidity (storage/operating)		< 95% RH (non-condensing) (0-95)
Internal Maintenance Temperature		25±5°C
Anti-corrosion grade		C4
Maximum working altitude		≤4000m
IP level		IP55
Operational environment		Outdoor (container)
Fire suppression system		Aerosol, combustible gas detection, supply and exhaust ventilation, explosion vent panel
Communication		Ethernet / CAN / RS485
System communication protocol		Modbus TCP/IEC104/IEC61850 (CAN Communication with the Control Unit / Ethernet Communication with EMS)
Certificates		IEC 62619/ IEC 63056/ IEC 61000/ IEC 62477-1/ UN 3536

1.3.2 System Components

Table Components of the container system

No.	Sub-parts	Components	Quantity	Remark
1	Battery container	20HC	1	Including distribution and lighting system
		Battery pack	48	1P104S(LFP 314Ah Cell), BMU
		Battery rack switchgear unit	12	Including BCMU, breaker, fuse, contactor, current sensor, etc.
		Power Distribution Cabinet	1	Including BAMS, disconnector, UPS
3	High Voltage Box	BCU, Circuit breaker, CT, Contactor	6	Control box of each two racks
4	BMS	BMS system	1	3-level architecture Including BMU, BCMU, BAMU
5	Liquid cooling system	Liquid cooling machine	1	Cooling capacity: 60kW (for 0.5C system)
		Liquid cooling pipes	1	Three-stage pipes
6	Fire suppression system	Aerosol	6	For container level
			96	For pack level
		Fire control panel	1	
		Smoke detector	2	Spot-type photoelectric
		Heat detector	2	Spot-type
		Gas detector	2	H_2
		Air intake fan	1	
		Explosion-proof exhaust fan	1	
		Water firefighting system	1	

1.4 Component Description

1.4.1 Battery Cell

Table Specification of battery cell

Item	Specification
Battery type	LFP

Capacity/ Ah		314
Nominal voltage/ V		3.2
Voltage range/ V		2.5 - 3.65
Rated energy/ Wh		1004.8
End-of-Dischar e		2.5 V T > 0°C; 2 V T ≤ 0°C
Initial internal resistance		0,18 mΩ ± 0,05 mΩ (AC, 1kHz)
Round-Trip Efficiency (RTE)		94%(0,5P)
Self-discharge		≤3%/month(25±2°C)
Charge/Discharge Current/A	Standard	157
Energy density/ Wh/kg		175
Life cycle (25°C, 90%DOD, 70%SOH)		8,000 (0.5P/0.5P)
Dimension (W * D * H)/ mm		(207.2±1.0) * (173.7±1.0) * (71.7±2.0)
Weight/ Kg		5.6±0.3
Operation temperature/ °C	Charge	0 - 55
	Discharge	-30 - 55
Certificates		UL 1973 / UL 9540A / IEC 62619 UN 38.3 / GB/T 36276 / ROHS / Bankability

1.4.2 Battery Pack

Table Parameters of battery pack

Item		Specification
Pack type		1P104S
Capacity/ Ah		314
Rated energy/ kWh		104.499
Configuration		1P104S
Nominal voltage/ V		332.8
Voltage range/ V		260 - 379.6
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 50
	Discharge	-20 - 50
BMS - Balancing Mode		Passive balancing
Efficiency		>93%
Standard charging power/current(0.5P,25±2°C)		52,25kW/157A
Standard discharge power/current (0.5P,25±2°C)		52.25kW/157A

Recommended ambient temperature	20 - 30℃
Charge/discharge efficiency （0,5P@25±3℃）	≥93%
Self-discharge/month	≤3.0%/month（Turning off auxiliary power）
Communication method	Daisy Chain
Balancing method	Passive
Auxiliary power system	24VDC
Self-power consumption	< 1W
Connection method	Quick connector
Dimension (W * D * H)/ mm	(790±5) * (2180±5) * (252±5)
Weight/ Kg	~690±10
IP level	IP67
Thermal management	Liquid cooling
Certificates	IEC 62619/63056/EN IEC 61000-6-2/61000-6-4

Figure The diagram of battery pack (Reference)

1.4.3 Battery Rack

Table Parameters of battery rack

Item	Specification
Rack type	1P416S
Battery type	LFP
Capacity/ Ah	314
Rated energy/ kWh	417.996 (0.5P@25±3℃)
Configuration	1P416S
Nominal voltage/ V	1331.2

Voltage range/ V		1164.8 – 1497.6
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 50
	Discharge	-30 - 50
Standard charging power/current(0.5P,25±2°C)		209kW/157A
Standard discharge power/current (0.5P,25±2°C)		209kW/157A
Recommended ambient temperature		20 - 30°C
Self-power consumption		<4W (Excluding liquid cooling power)
Connection method		Quick connector
Thermal management		Liquid cooling
IP level		IP67
Dimensions(WxDxH)		790x1140x2333 mm（Tolerance:±3mm）
Certificates (All finished)		IEC 62619/63056/EN IEC 61000-6-2/61000-6-4